

Modulus and argument

Basic skills

Q1. a) $3+4i$

$x=3, y=4$

$$|z| = \sqrt{x^2 + y^2}$$

$$|3+4i| = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$$

b) $3-4i$

$$|3-4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9+16} = 5$$

c) $|5i-1| = \sqrt{(-1)^2 + 5^2} = \sqrt{1+25} = \sqrt{26}$

Q2. a) $3+4i$

$x=3, y=4$

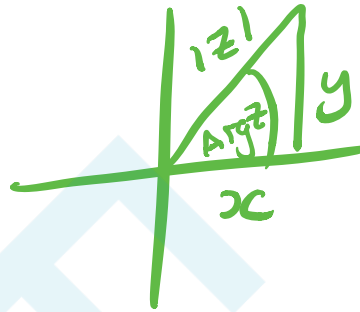
$$\text{Arg } z = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\text{Arg}(3+4i) = \tan^{-1}\left(\frac{4}{3}\right) = 0.927$$

b) $\text{Arg}(1-9i) = \tan^{-1}\left(\frac{-9}{1}\right) = -1.46$
 $x=1, y=-9$

c) $\text{Arg}(10i+1) = \tan^{-1}\left(\frac{10}{1}\right) = 1.47$
 $y=10, x=1$

Q3. modulus is distance from 0 on argand diagram $(0,0)$ to (x,y) and Argument is angle to positive real axis



Competency skills

Q4. a) $z = -3 + 4i$



b) $|z| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

$\text{Arg } z = \tan^{-1}\left(\frac{4}{-3}\right) = -0.927$

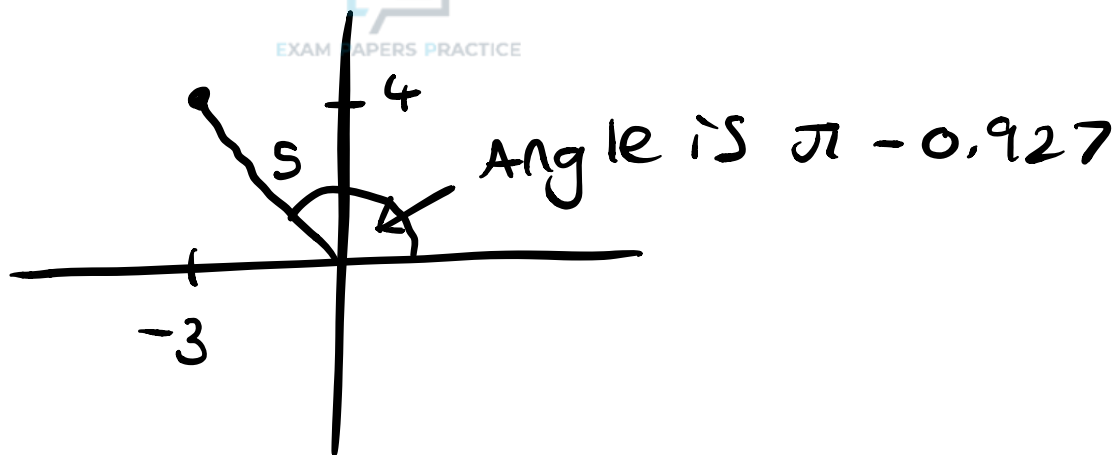
However $x < 0$ so careful,
look at diagram to see



so

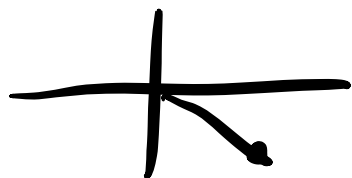
$\text{Arg } z = \pi - 0.927$

c)



Q5. a) $|9 - 7i| = \sqrt{9^2 + 7^2} = \sqrt{130}$

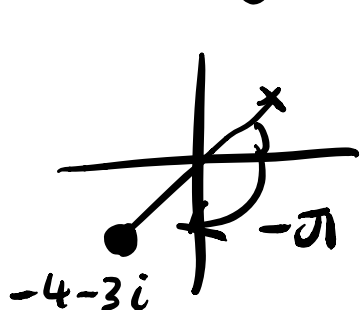
$\text{Arg}(9 - 7i) = \tan^{-1}\left(\frac{-7}{9}\right) = -0.661$



no need to alter
as $x > 0$

b) $|-4 - 3i| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$

$\text{Arg}(-4 - 3i) = \tan^{-1}\left(\frac{-3}{-4}\right) = 0.644$



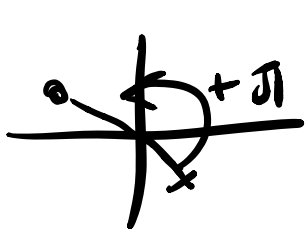
$x < 0$
 $y < 0$

so
 $\text{Arg}(-4 - 3i) = 0.644 - \pi$

so $-\pi$

$$c) |3i - 4| = \sqrt{4^2 + 3^2} = 5$$

$$\text{Arg}(3i - 4) = \tan^{-1}\left(\frac{3}{-4}\right) = -0.644$$



$x < 0$
 $y > 0$
 so $+\pi$

so

$$\text{Arg}(3i - 4) = \pi - 0.644$$

$$d) \left| \frac{4}{3} - i \right| = \sqrt{\left(\frac{4}{3}\right)^2 + 1^2} = \sqrt{\frac{16}{9} + 1}$$

$$= \sqrt{\frac{25}{9}} = \frac{5}{3}$$

$$\text{Arg}\left(\frac{4}{3} - i\right) = \tan^{-1}\left(\frac{-1}{\frac{4}{3}}\right) = \tan^{-1}\left(-\frac{3}{4}\right)$$

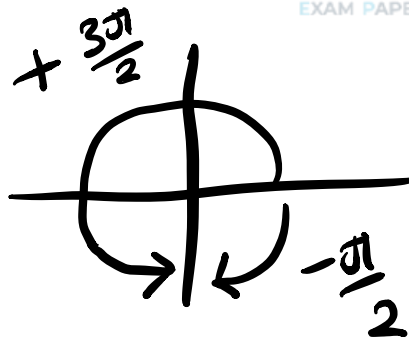
$$= -0.644$$

$$e) -\frac{1}{3}(i - 3) = 1 - \frac{1}{3}i$$

$$\left| 1 - \frac{1}{3}i \right| = \sqrt{1^2 + \left(\frac{1}{3}\right)^2} = \sqrt{\frac{10}{9}} = \frac{\sqrt{10}}{3}$$

$$\text{Arg}\left(-\frac{1}{3}(i - 3)\right) = \tan^{-1}\left(\frac{-\frac{1}{3}}{1}\right) = -0.322$$

Q6.



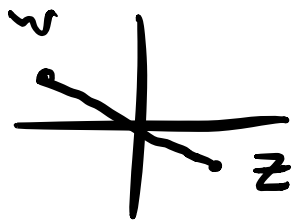
so $-\frac{\pi}{2}$ is
equivalent
for argument

Advanced skills

Q7. $z = 4 - i$ $w = -3 + i$

$$z - w = 4 - i - (-3 + i)$$

$$= 7 - 2i$$



$$|z - w| = \sqrt{7^2 + 2^2} = \sqrt{53}$$

Q8. $z = x + yi$ $z^* = x - yi$

$$|z| = \sqrt{x^2 + y^2}$$

$$|z^*| = \sqrt{x^2 + (-y)^2}$$

$$= \sqrt{x^2 + y^2}$$

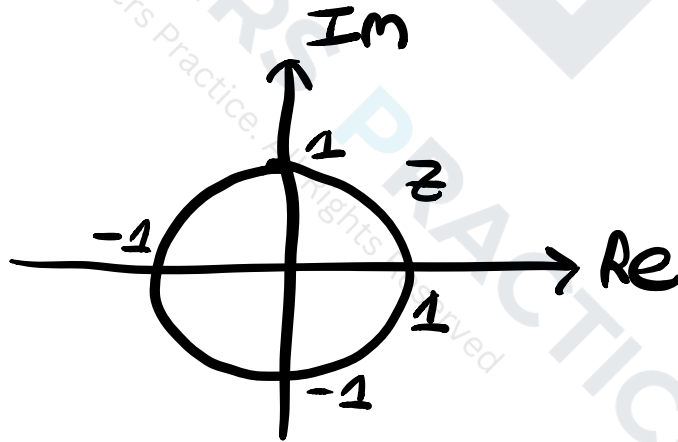
so $|z| = |z^*|$

so correct

Q9.

$$\begin{aligned} z z^* &= (x+yi)(x-yi) \\ &= x^2 + yxi - yxi - y^2 i^2 \\ &= x^2 + y^2 = \sqrt{x^2 + y^2}^2 = |z|^2 \end{aligned}$$

Q10. If $|z|=1$ the complex number is such that its distance from 0 is 1 so

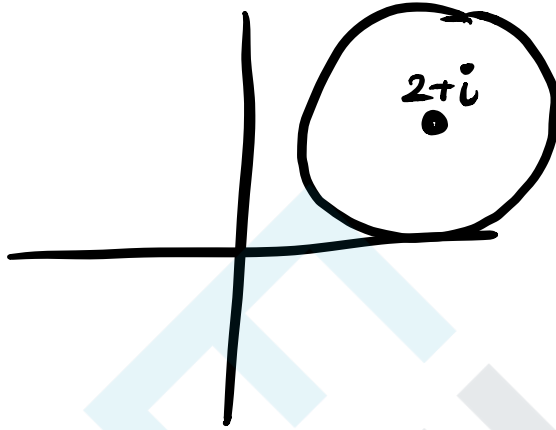


e.g. $z = 1, i, -i, -1$

& $z = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$



Q11. similarly to Q10. a distance
of 1 from $(2+i)$



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